CSCI 5526

Interpreting ML Models in Healthcare

Announcements

- MP3 due 10/23.

- Finalize project groups by Friday!
  - Next milestone: project proposals (due 10/30)
  - https://z.umn.edu/ml4h_project_proposal_template
  - https://z.umn.edu/ml4h_project_datasets

Student Questions

How does the MaxPooling layer in a Convolutional Neural Network help preserve important features, such as tumors or lesions, while reducing the dimensionality of medical images? What are the potential risks of losing critical diagnostic information during pooling?

Max-pooling will preserve high-contrast/sharp features. Tumors and lesions often manifest as high-contrast regions or sharp intensity spikes w.r.t. the background. These will have high activation values in a CNN and will pass through max-pool layers.

However, there is a significant risk if the important features have heterogeneous shapes/attributes.

Student Questions

Which padding method is used most often in medical imaging models?

Same padding (aka zero padding) is most common as it preserves the dimensions.

For temporal medical data, do people use RNNs or LSTMs? Or even hybrid model like Conv-LSTMs?

Yes, if there is a need to learn morphological patterns before the temporal patterns (such as in signal data), adding conv layers before RNN/LSTM layers is useful.

Student Questions

How specific should we get for the group project proposal? If I want to use convolutional neural networks, should I specify planned architectures? What if we aren't sure yet?

Try to be as specific as possible, especially if you are using pretrained models.

If you want to experiment with a custom architecture, then it is OK to keep the description a bit general.

But please do think about the input size, dataset size, and other relevant aspects to narrow down the architectural choice.

Student Questions

How do you determine # of convolution layers and/or dimensions to reduce to in each layer?

If you have a large dataset:

classification: 5-10 convolutional layers

segmentation: 10-20 convolutional layers with skip connections

For a small dataset: 3-5 convolutional layers to reduce overfitting risk

Generally, reduce spatial dimensions progressively (e.g., halving at each step), while doubling feature channels to capture more abstract representations.

For segmentation tasks, if you downsample too much and you lose fine boundary details.

For classification, it's okay to reduce more aggressively as long as you preserve semantic meaning.

Student Questions

How to choose appropriate architecture for the dataset we have? Are there any key considerations?

Dataset size:

>10k images: train a deep model from scratch

1k-10k images: pretrained model + finetuning

<1k images: transfer learning or custom architectures

For Segmentation

- U-Net: Efficient for small datasets, captures fine features.
- nnU-Net: Auto-adapts architecture to your dataset. Strong default.
- Attention U-Net: Highlights relevant features (good for small/ambiguous lesions).

For Classification

- ResNet-18/34: Simple, low risk of overfitting.
- DenseNet: Good at feature reuse, often better for small datasets.
- EfficientNet: Highly optimized, but sensitive to training settings.

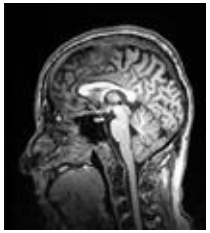
Outline

- Why do we need interpretability
- Interpretability vs explanations
- Types of explanations
- Explainability methods
  - Classification of methods
  - CAM, LIME, SHAP
  - Counterfactual methods
- Some specific case-studies

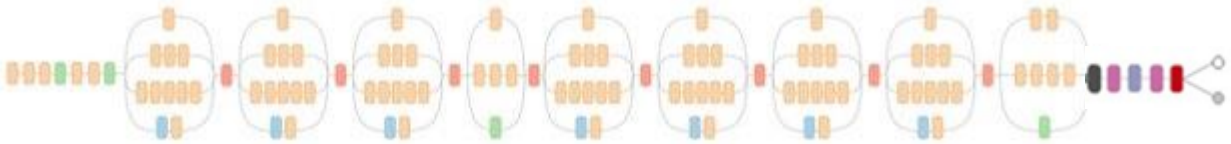
- A good resource: https://christophm.github.io/interpretable-ml-book/

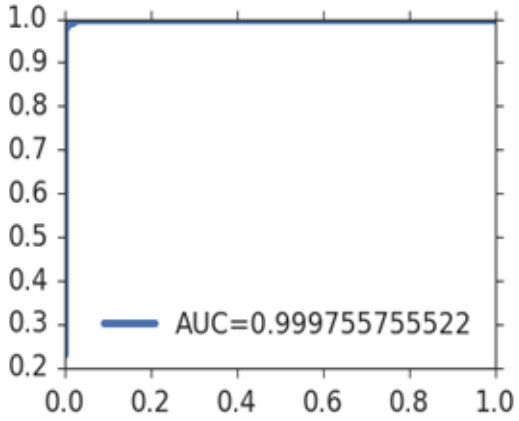
A Motivating Result:

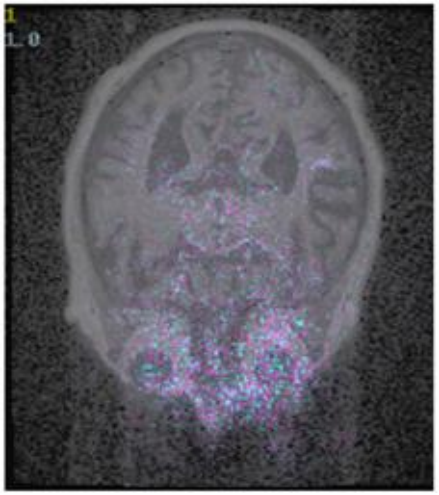
Prediction of biological sex using structural MRI

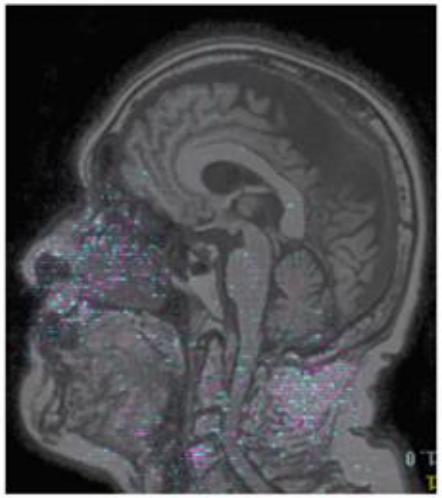


Pretrained Inception 3D feature extractor









Question

- What is interpretability?

Interpretability is the degree to which a human can understand the cause of a decision

Interpretability is the degree to which a human can consistently predict the model's result.

Question

- Why do we need interpretability?

Debugging

- Why did the system make a wrong prediction in this case?
- What does it actually learn? What kind of data would make it better?
- How reliable/robust is it?
- Understanding edge cases

Most common use case in practice according to recent study (Bhatt et al. "Explainable machine learning in deployment." In Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency, pp. 648-657. 2020.)

Auditing

- Understand safety implications
- Ensure predictions are based on objective criteria and reasonable rules Inspect fairness properties
- Reason about biases and feedback loops

Discovery

- Does a feature contribute to predictions?
- Is there a structure in the feature space that corresponds to differences in classes?
- Which feature is the prediction most sensitive to?

Interpretability in Healthcare

- High stakes
  - Errors can have life threatening consequences
- Deciding interventions
  - Would be useful to know how to treat a condition
- Equitable healthcare
  - Want to make sure our models are fair and unbiased
- Discover new knowledge
  - Novel therapeutic targets or disease subtypes

Types of Interpretability

Intrinsic

Some models are inherently interpretable.

Post-hoc (explainability)

Some models are complex and hard to interpret. We apply external techniques to explain those models.

Intrinsic Interpretability vs Post-hoc Explainability

Models simple enough to understand (e.g., short decision trees, sparse linear models)

1. *Congestive Heart Failure*	1 point		⋯
2. *Hypertension*	1 point	+	⋯
3. *Age* $\geq$ *75*	1 point	+	⋯
4. *Diabetes Mellitus*	1 point	+	⋯
5. *Prior Stroke or Transient Ischemic Attack*	2 points	+	⋯
ADD POINTS FROM ROWS 1–5	**SCORE**	=	⋯

SCORE	0	1	2	3	4	5	6
STROKE RISK	1.9%	2.8%	4.0%	5.9%	8.5%	12.5%	18.2%

Explanation of black-box models local or global

You are not at risk of cardiovascular disease because your BMI is less than 30.

People with BMI less than 30 are typically not at risk of cardiovascular disease.

Terminology

- Accepted terminology:
  - Interpretable models: Intrinsically interpretable models
  - Explainability: Post-hoc explanations

- Interpretability: property of a model
- Explainability: ability to explain the workings/predictions of a model
- Explanation: justification of a single prediction

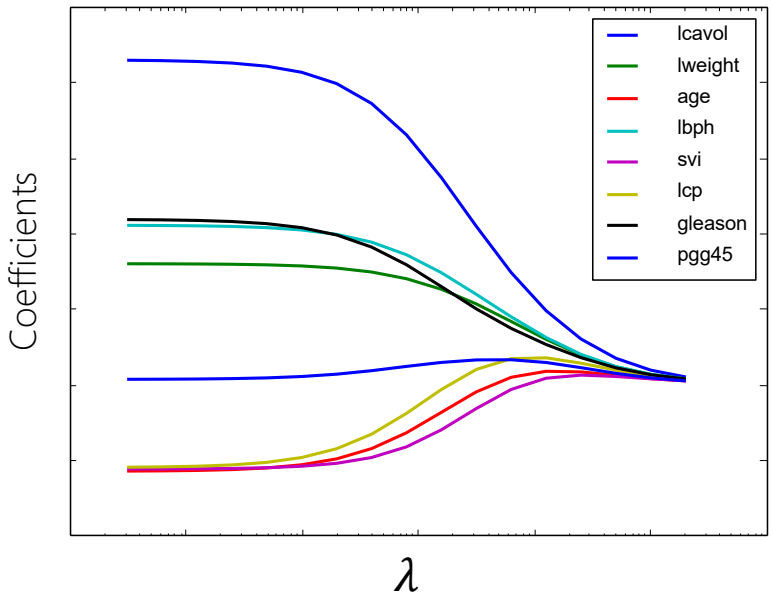
- Interpretability and explainability are often used inconsistently or interchangeably

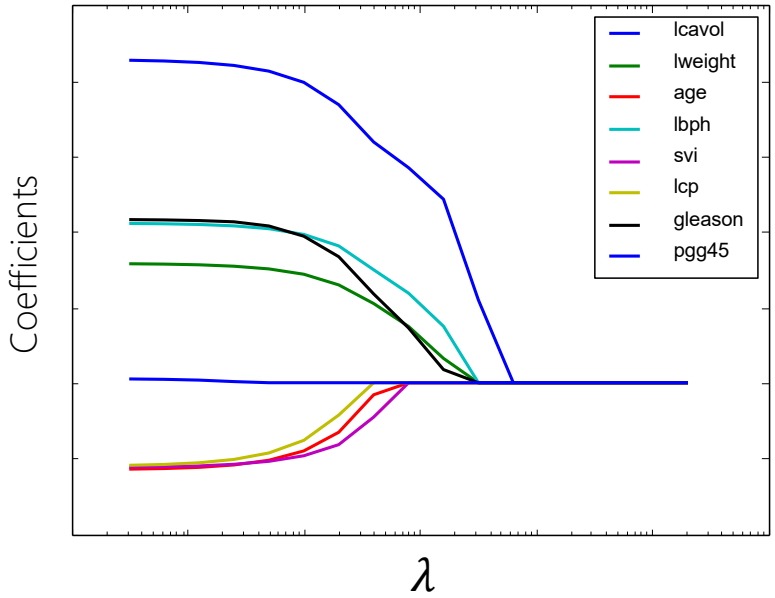
Interpretable Models

- Linear/logistic Regression
- Generalized linear models
- Decision Trees
- k-nearest neighbors

Ridge and LASSO Regression

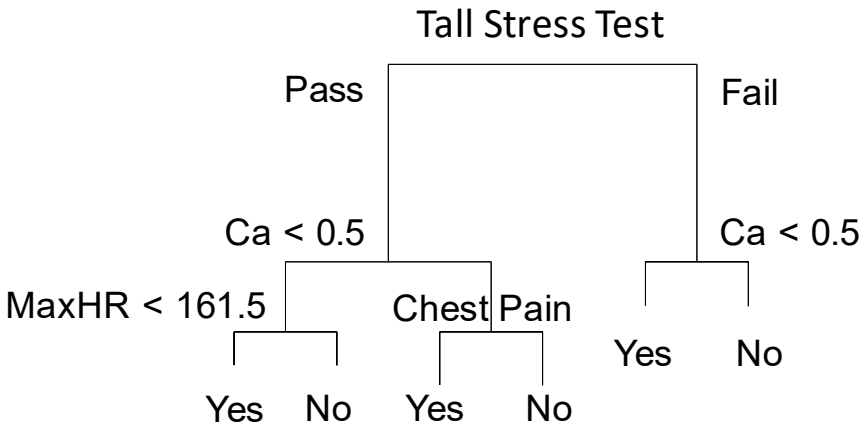
- Task: predict prostate-specific antigen (PSA) levels in patients who underwent radiation therapy for prostate cancer.
- The goal is to predict the log of PSA (lpsa) from a number of measurements including log cancer volume (lcavol), log prostate weight (lweight), age, log of benign prostatic hyperplasia amount (lbph), seminal vesicle invasion (svi), log of capsular penetration (lcp), Gleason score (gleason), and percent of Gleason scores 4 or 5 (pgg45).





Decision Tree

- Intuitive way of making a decision based on series of simple rules



Example: Heart Disease Prediction

Source: Elements of Statistical Learning Book

Interpretable Models

- Pros
  - Simple
  - Easy to understand
  - Inherent interpretability

- Cons
  - Too simple
  - Do not have the capacity to learn complex interactions

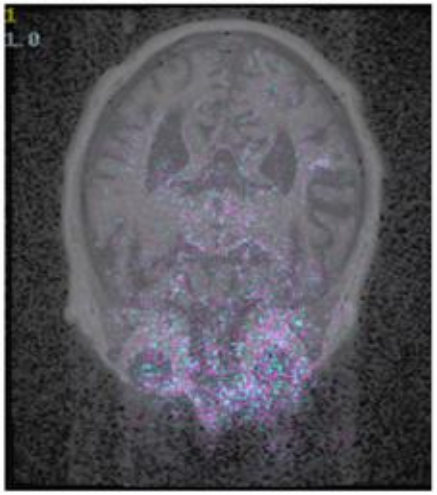
Post-hoc Explanations

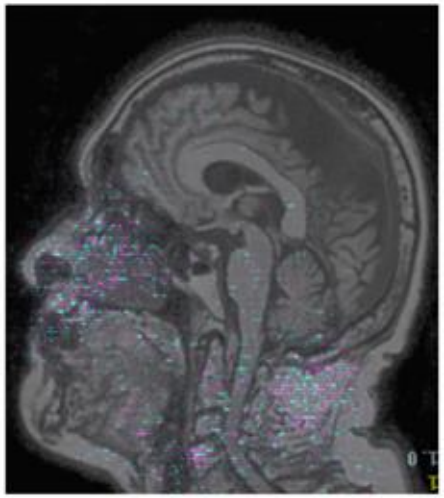
- General idea
  - Explain an inherently uninterpretable model
- Two types
  - White-box – we know the model
  - Black-box – know nothing about the model
- Another classification
  - Local – explain individual predictions
  - Global – explain the overall model

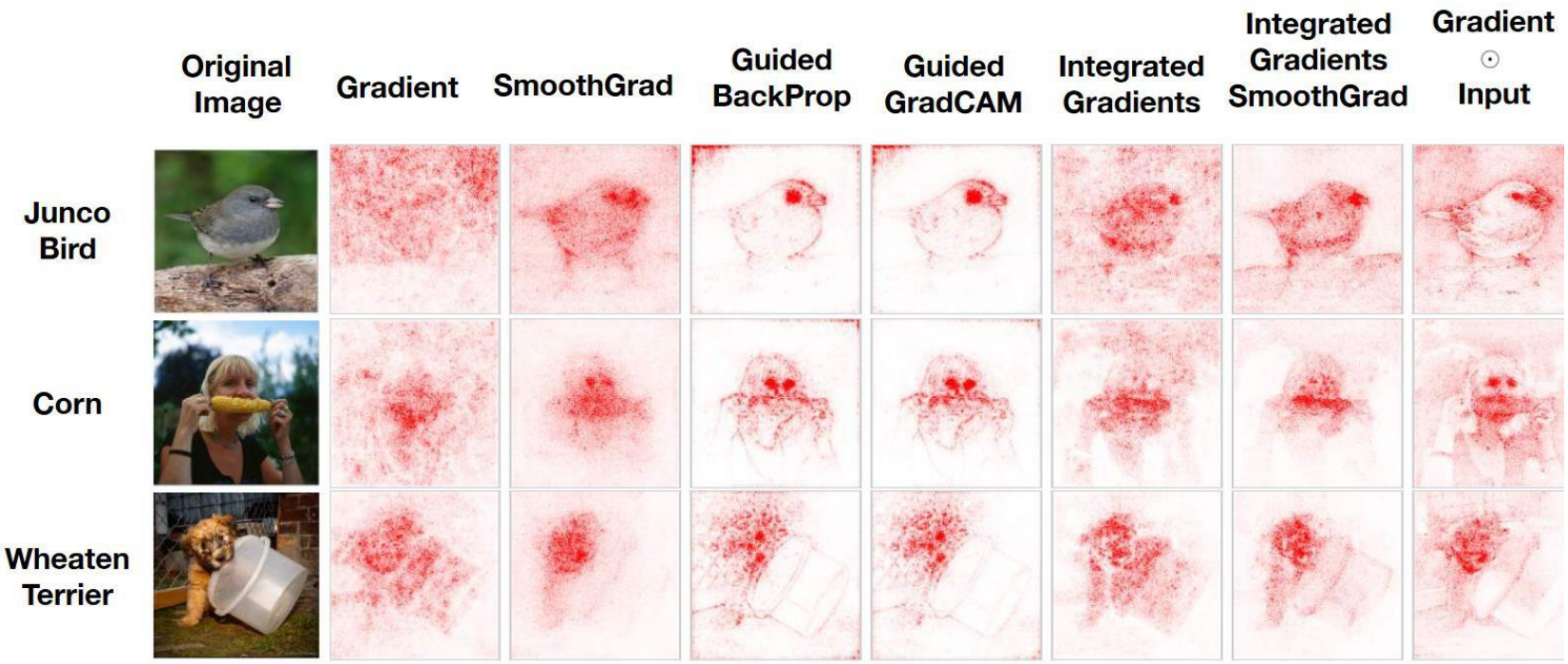
Some locally Interpretable, Post hoc methods

Saliency Based Methods

- Heatmap based visualization
- Need differentiable model in most cases
- Normally involve gradient







[Adebayo et al 2018]

Saliency Example - Gradients

$$f(x) \quad : R^d \rightarrow R$$

$$E(f)(x) = \frac{df(x)}{dx}$$

Why do we use gradient?

Saliency Example – Leave-one-out

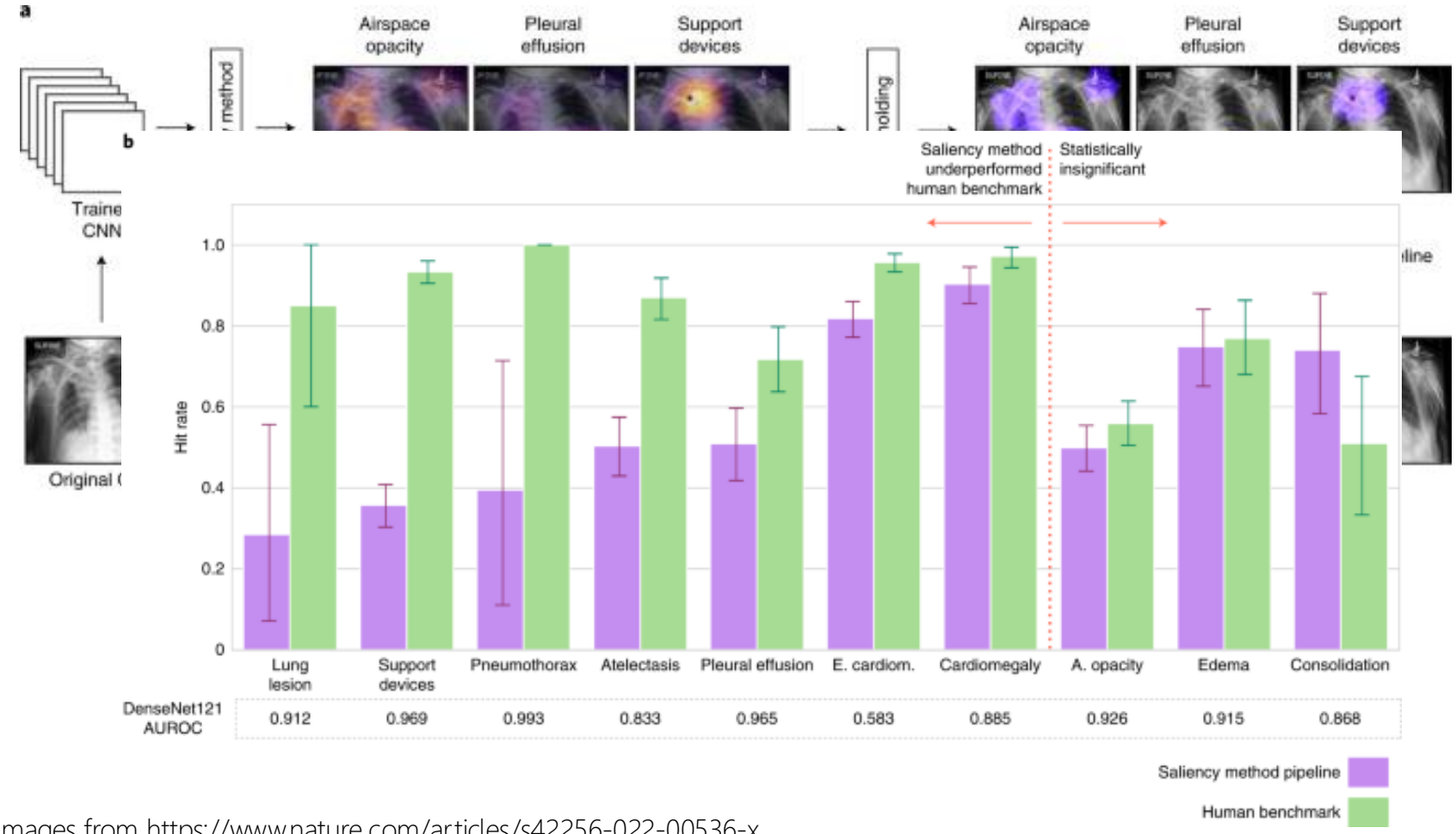
$$f(x): R^d \rightarrow R$$

$$E(f)(x)_i = f\ \ (x) - f(x\backslash \mathrm{i})$$

How to remove a feature?

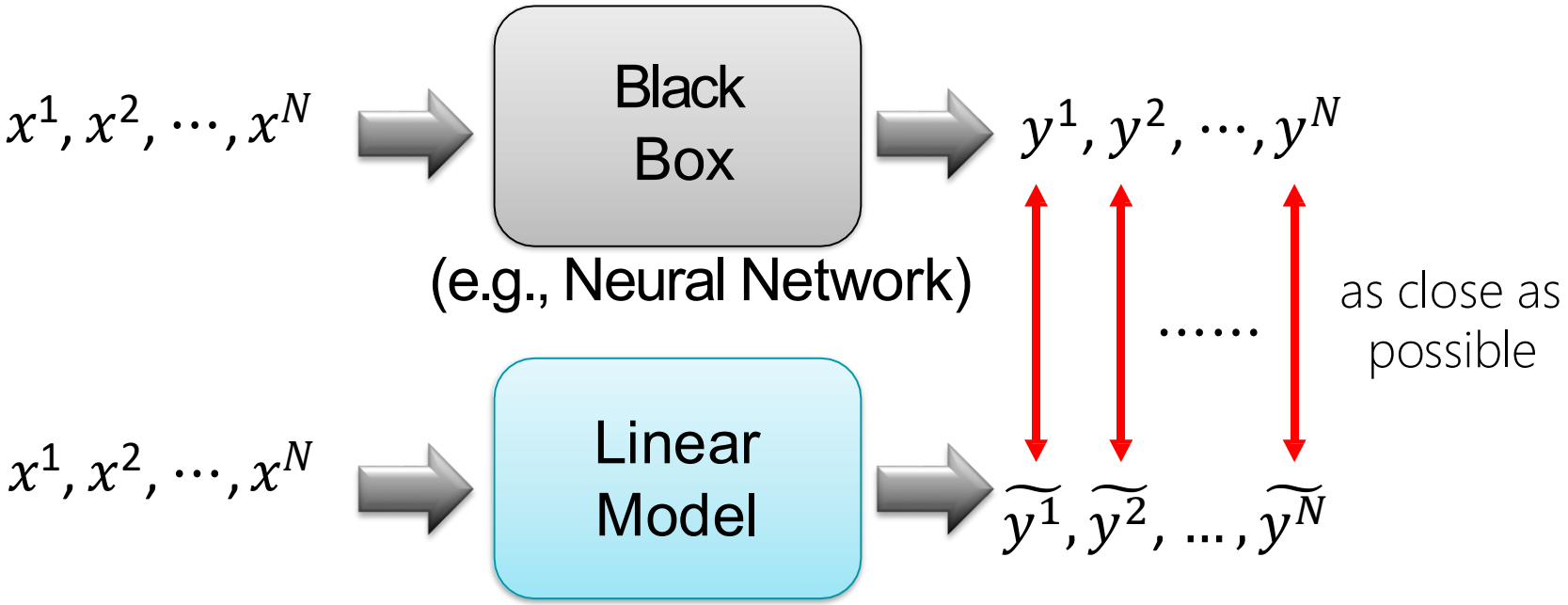
1. Zero out pixels in image
2. Remove word from the text
3. Replace the value with population mean in tabular data

Saliency: Does it work?



Images from https://www.nature.com/articles/s42256-022-00536-x

LIME – locally interpretable model agnostic



Can't do it globally of course, but locally? Main Idea behind LIME

(Image Credit – Hung-yi Lee)

Intuition behind LIME

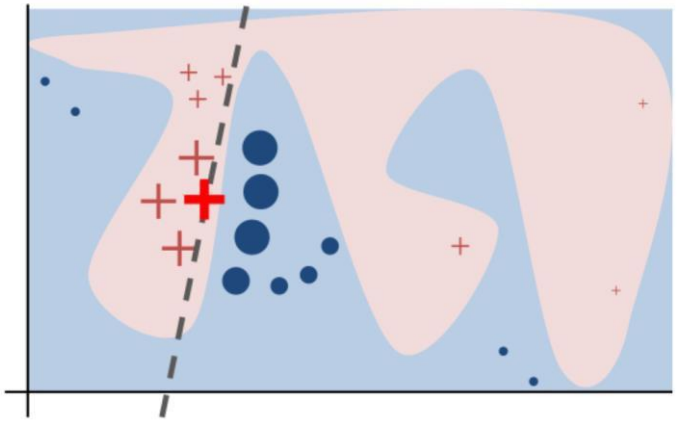
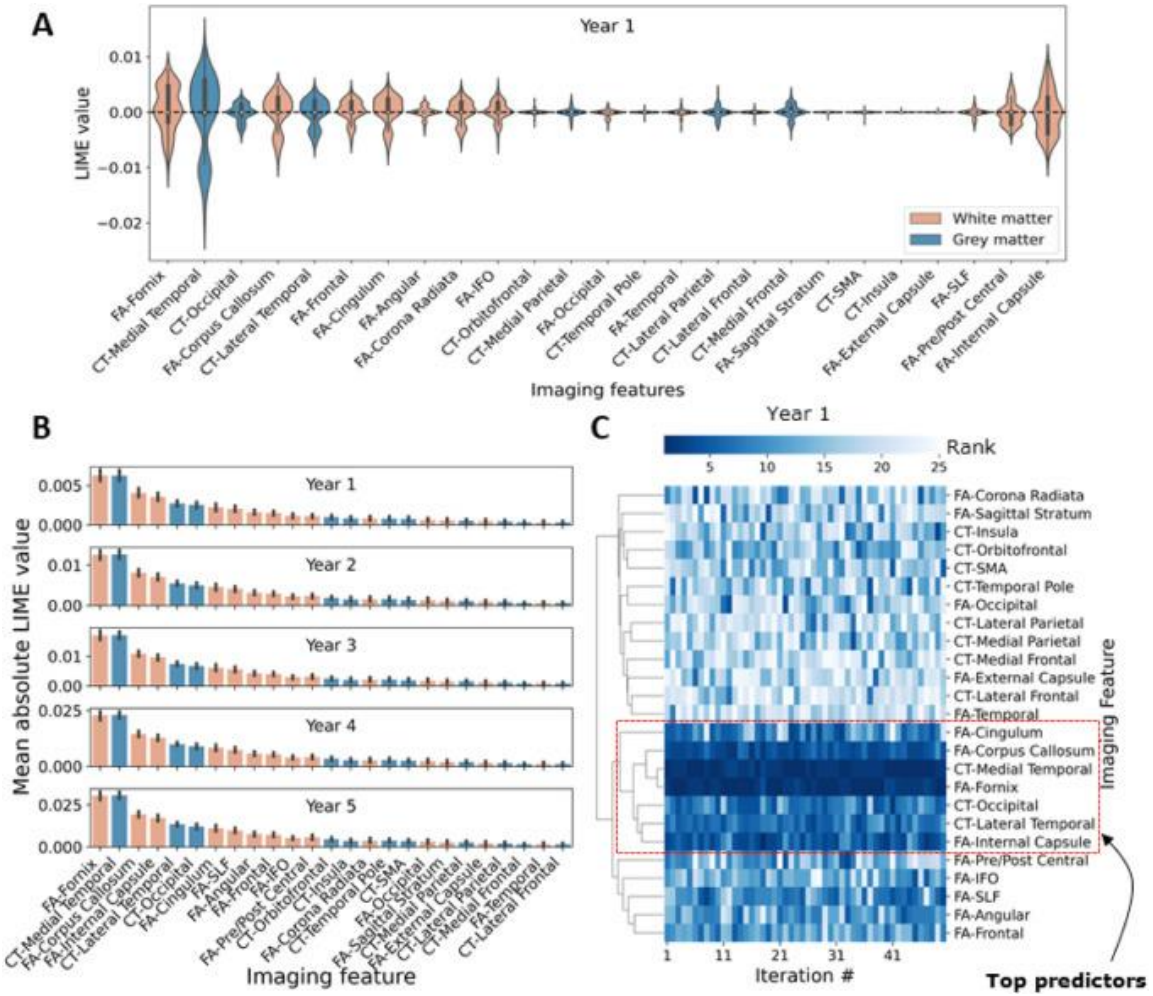


Figure 3: Toy example to present intuition for LIME. The black-box model's complex decision function f (unknown to LIME) is represented by the blue/pink background, which cannot be approximated well by a linear model. The bold red cross is the instance being explained. LIME samples instances, gets predictions using f, and weighs them by the proximity to the instance being explained (represented here by size). The dashed line is the learned explanation that is locally (but not globally) faithful.

[Ribeiro et al 2016]

LIME Example



Images from https://www.sciencedirect.com/science/article/pii/S1053811922001495

SHAP – SHapely Additive Explanations

- Inspired from Shapely values.
  - An approach used to fairly distribute profits in corporate environments.
  - Due to Lloyd Shapely (won the 2012 Nobel Prize for economics).

- SHAP uses Shapley values to explain individual predictions of an ML model.

Example

- A company with two employees: Alice and Bob

- No employees, no profit [v({}) = 0]
- Alice alone makes 20 units of profit [v({Alice}) = 20]
- Bob alone makes 10 units of profit [v({Bob}) = 10]
- Alice and Bob make 50 units of profit [v({Alice, Bob}) = 50]

- What should the bonuses be?

Answer

- No employees, no profit [v({}) = 0]
- Alice alone makes 20 units of profit [v({Alice}) = 20]
- Bob alone makes 10 units of profit [v({Bob}) = 10]
- Alice and Bob make 50 units of profit [v({Alice, Bob}) = 50]

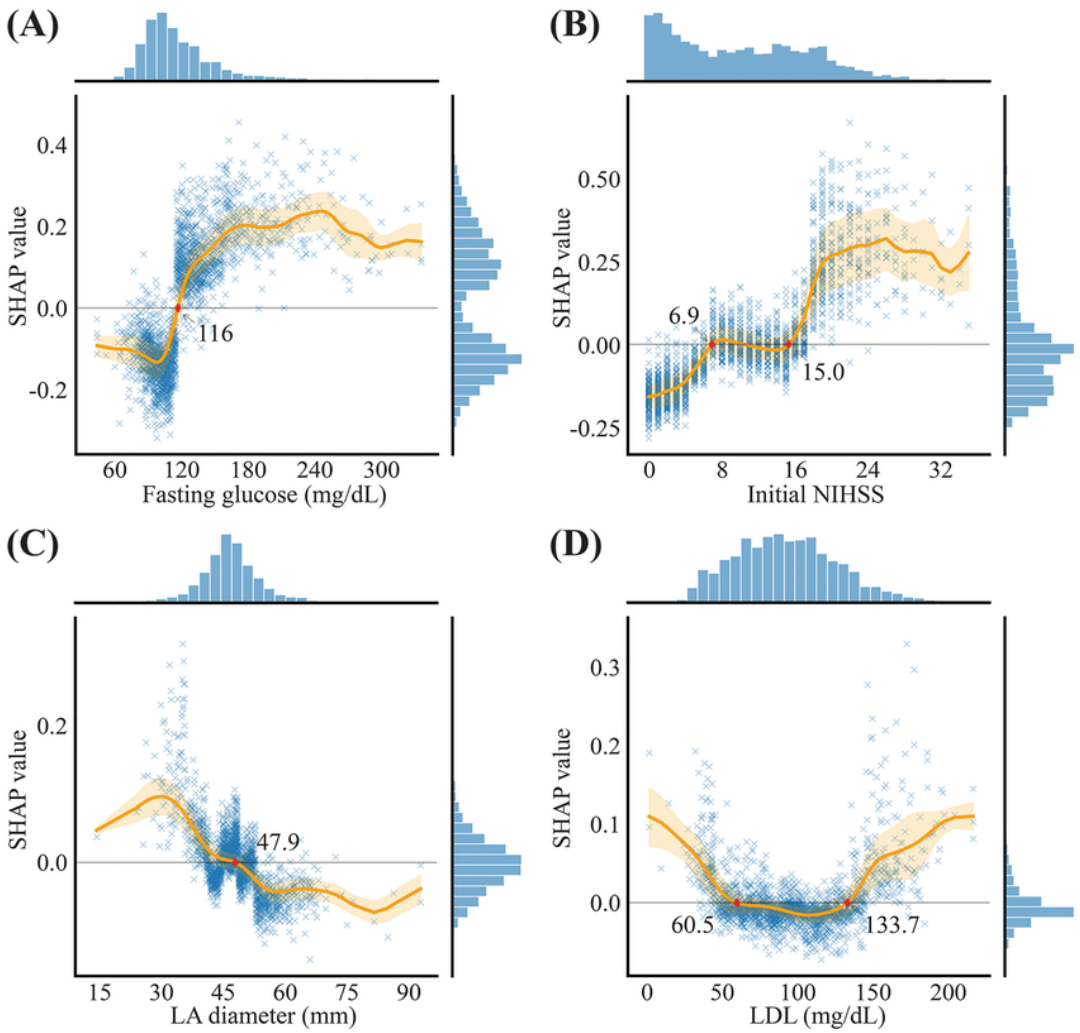
Permutation	Marginal for Alice	Marginal for Bob
Alice, Bob	20	30
Bob, Alice	40	10
Shapley value	***30***	***20***

Application to ML

- Consider features as players
- Consider model behavior as profit
  - E.g., the prediction, the loss, etc.
- Then, use Shapley values to quantify each feature's impact

Example

Early neurological deterioration prediction in atrial fibrillation-related stroke.



Kim, Seong-Hwan, et al. "Interpretable machine learning for early neurological deterioration prediction in atrial fibrillation-related stroke." Scientific reports 11.1 (2021): 20610.

Counterfactual Explanations

- Asking questions like:
  - What would happen if we change the input feature values?
  - What kind of modifications of the input will change the prediction?
  - What happens if we remove one or more training samples during model training?

Adversarial Counterfactual Explanations

- Consider the stroke prediction example.
- Suppose the model predicts that for one individual the risk is 90%.

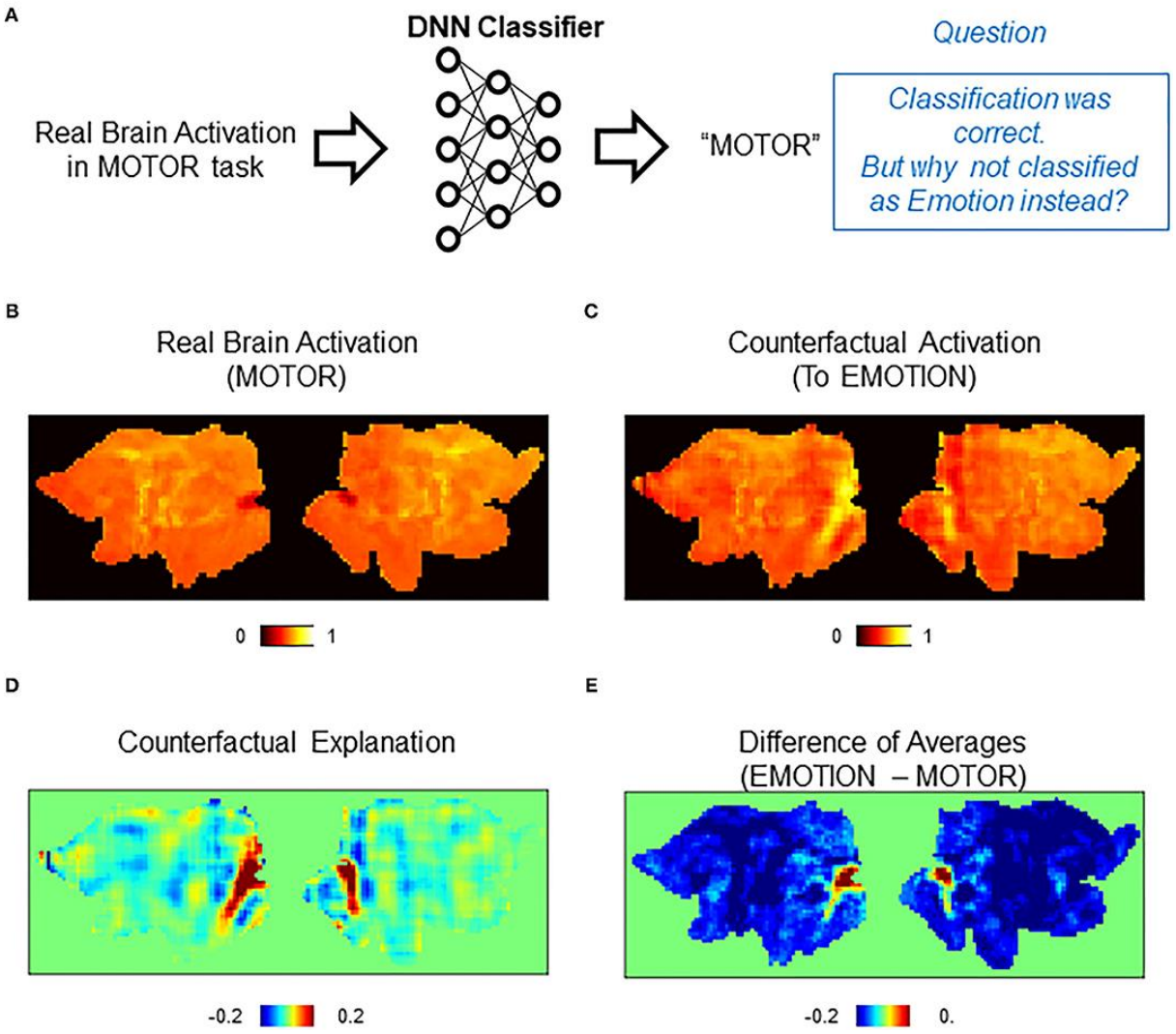
- A counterfactual explanation would be,
  - If you reduce your BMI to 25 your stroke risk would become 10%.

- Mathematically,

$$argmin_{x'}\ d(x, x')$$
$$s.t.\ f(x') = c'$$

Example

- Counterfactual explanations in fMRI-based brain activity classification.

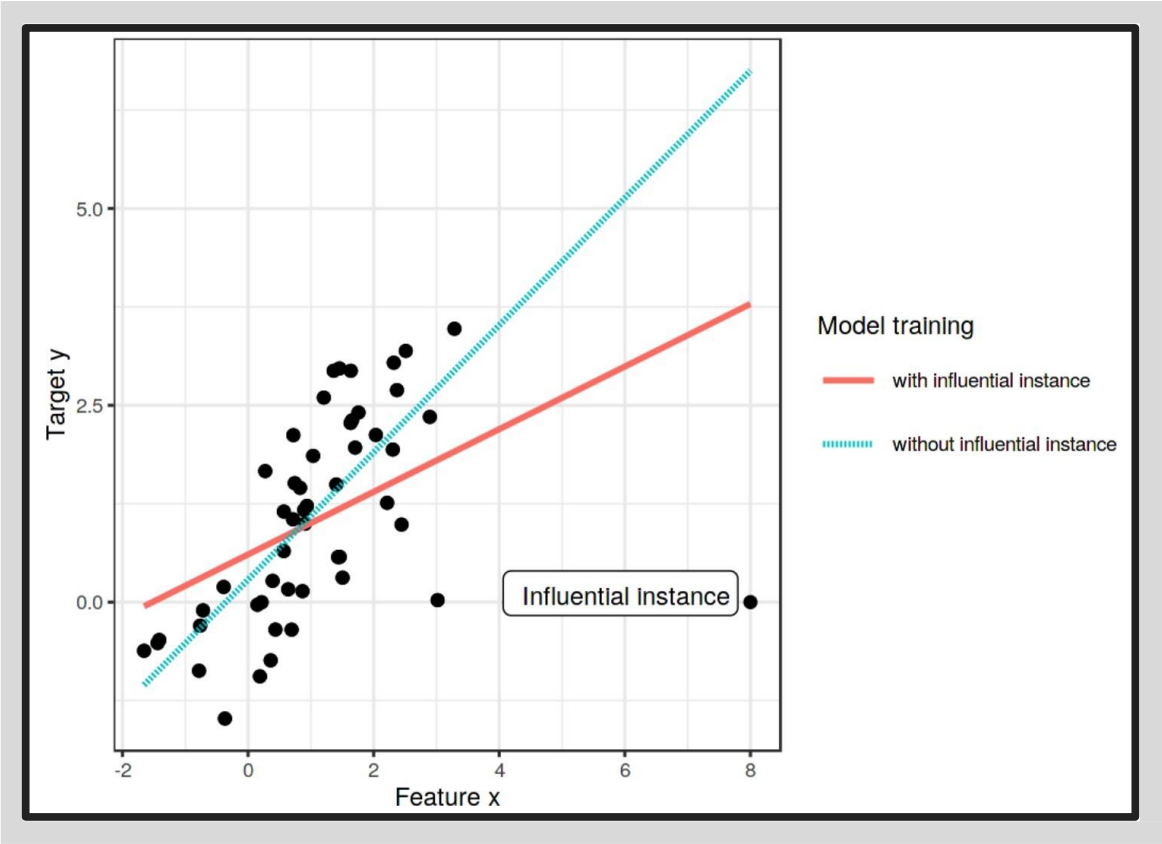


Matsui, Teppei, et al. "Counterfactual explanation of brain activity classifiers using image-to-image transfer by generative adversarial network." Frontiers in Neuroinformatics 15 (2022): 802938.

Influential Instances

- What data most influenced the training? Is the model skewed by few outliers?
  - Training data with n instances
  - Train model f with all n instances
  - Train model g with n – 1 instances
  - If f and g differ significantly, omitted instance was influential
    - Difference can be measured e.g., in accuracy or difference in parameters

Example



Source: Christoph Molnar. "Interpretable Machine Learning." 2019

Why are counterfactual explanations important in healthcare?

Discussion Questions

- Are there applications of ML for which you don't need interpretability?

- How to evaluate explanation approaches?
  - How to tell one is better than another

Exit pass

https://z.umn.edu/ml4h_exitpass

